Kumudini Gandesiri

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Summary

Hardware validation and bring-up professional with a B.E. in Electronics and Communications Engineering, an M.S. in Management Information Systems, and hands-on semiconductor lab experience at AMD and Apple. Debugs PCIe link and board-level failures, analyzes test data with JMP, Python, and SQL, and presents root-cause findings to engineering teams. Builds AI agents and Python automation that speed up analysis and reduce manual work.

Skills

Interfaces & Memory: PCIe (link training, enumeration), DRAM timing, I2C, SPI, UART, FPGA/JTAG
Debug & Validation: Linux (bash, lspci, dmesg, journalctl), Vivado ILA, oscilloscope, DMM, SEM, curve tracer
Data Analysis: Python (pandas, matplotlib), SQL, Snowflake, JMP, Excel, Power BI, Streamlit, SPC, Cpk
AI & Automation: Google Cloud Generative AI, LLM agents, Python and shell automation, KPI and exception reporting
Technical Communication: Test documentation, defect reports, design documentation, root-cause presentations

Experience

Hardware Lab Technician (ASIC Bring-Up) May 2026 – Present
AMD, San Jose, CA (Contract)

- Bring up new ASIC platforms in a fast-paced lab, focusing on PCIe link-up on dedicated test boards and lab testers.
- Analyze PCIe link training data and lspci, dmesg, and journalctl output to root-cause marginal links and errors.
- Debug failing boards with Vivado ILA signal captures to isolate dead components, broken traces, and solder defects.
- Run automated I2C, SPI, and UART validation routines on FPGA test rigs while monitoring power rails through ADCs.
- Write shell and Python scripts to automate test runs and log collection, and document test results and board changes.

Hardware Lab Technician (Hardware Validation) Jan 2025 – Feb 2026
Apple, Santa Clara, CA (Contract)

- Analyzed electrical test data in JMP, Python, SQL, and Excel, using distributions and outlier screening to find trends, marginal behavior, and component-level failures.
- Monitored test parameters with control limits, Cpk, and SPC charts to flag parameters drifting toward spec limits.
- Performed electrical and physical failure analysis with SEM, curve tracer, and optical microscopy, and presented root-cause findings to engineering teams.
- Debugged system bring-up in Linux by checking kernel messages, device nodes, and hardware interfaces.
- Automated data collection and log parsing with Python and shell scripts, reducing manual analysis work.

Assembly Test Technician Jul 2022 – Jan 2023
Team Engineers Pvt Ltd, Hyderabad, India (Contract)

- Tested circuit boards on the production line with oscilloscopes, DMMs, and custom fixtures, and wrote defect reports.

Projects

DRAM Spec-Limit Screening Agent | *Google Cloud Generative AI, Python*

- Built an AI agent that screens open-source DDR2–DDR5 benchmark data against spec limits and reports pass rates.

Fab Production Scheduling & Capacity Optimization System | *Python, SQL, Snowflake, Streamlit, Power BI, Git*

- Built Streamlit and Power BI dashboards on simulated wafer-lot data to track throughput, cycle time, and WIP.
- Automated data validation and exception reports, and documented requirements, database design, and testing.

Wafer Defect Pattern Analysis & Excursion Monitoring (In Progress) | *Python, scikit-learn, Streamlit*

- Analyzing 811K open-source wafer maps for defect trends, and developing SPC excursion charts and an ML classifier.

Education

M.S., Management and Information Systems, Northwest Missouri State University, Maryville, MO 2024
B.E., Electronics and Communications Engineering, Jawaharlal Nehru Technological University, Hyderabad 2022
Relevant Coursework: VLSI Design, Digital Circuits, Python Programming, Statistics and Analytics, Database Management